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Jianpi Huayu Decoction suppresses cellular senescence in colorectal cancer via p53-p21-Rb pathway: Network pharmacology and in vivo validation

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ABSTRACT

Ethnopharmacological relevance: Jianpi Huayu Decoction (JHD) is an herbal prescription in traditional Chinese medicine based on Sijunzi Decoction to treat patients with colorectal cancer (CRC). Its effects on the inhibition of CRC cell proliferation and tumor growth are promising; however, its multicomponent nature makes a complete understanding of its mechanism challenging.
Aim of the study: To explore the therapeutic targets and underlying molecular pathways of JHD in CRC treatment using network pharmacology techniques and in vivo validation.
Materials and methods: The active ingredients and targets of JHD were identified, compound-target interactions were mapped, and enrichment analyses were conducted. We identified the hub targets of JHD-induced cellular senescence in CRC. The binding affinities between compounds and targets were evaluated through molecular docking. Subsequently, we conducted bioinformatic analyses to compare the expression of hub targets between colorectal tissue and normal tissue. Finally, in vivo experiments were carried out utilizing a xenograft model to assess the effects of JHD on cellular senescence biomarkers.
Results: Network pharmacology revealed 129 active ingredients in JHD that were associated with 678 targets, leading to the construction of compound-target and target-pathway networks. Enrichment analyses highlighted key pathways including cellular senescence. Based on this, hub targets associated with cellular senescence were determined and validated. Molecular docking indicated favorable interactions between the active components and hub targets. Gene expression and survival analysis in CRC tissue were consistent with the potential roles of hub genes. Animal experiments showed that JHD triggered cellular senescence and suppressed the growth of CRC by regulating the p53-p21-Rb signaling pathway.
Conclusions: This research adopted network pharmacology, bioinformatics, and animal experiments to unveil that JHD induces cellular senescence in CRC by influencing the p53-p21-Rb pathway and senescence-associated secretory phenotypes, highlighting its potential as a CRC treatment.

1. Introduction

Colorectal cancer (CRC) is a prevalent malignancy of the digestive system. According to GLOBOCAN 2020 (Sung et al., 2021), colorectal cancer ranks as the third most common cancer and the second leading cause of cancer-related deaths worldwide. In China, CRC ranks among the top five cancers, with approximately 408,000 newly diagnosed cases leading to around 195,000 deaths (Zheng et al., 2022). Unfortunately, most patients with CRC are diagnosed at advanced stages, leading to limited opportunities for curative surgery due to extensive local infiltration and distant metastases. Despite significant advancements in systemic treatments such as cytotoxic chemotherapy and biological therapy for metastatic CRC (Biller and Schrag, 2021), the prognosis for patients with CRC remains unfavorable.

Cellular senescence is an irreversible halt in the cell cycle induced by factors such as DNA damage, oxidative stress, and telomere attrition (Calcinotto et al., 2019). Senescent cells display heightened expression of cycle-regulating proteins such as p53, p21, and p16. Additionally, they release a diverse array of factors comprising the senescence-associated secretory phenotype (SASP), encompassing IL-6,

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IL-8, IL-1, VEGF, and MMP, which can affect both themselves and neighboring cells through autocrine and paracrine pathways (Gorgoulis et al., 2019). Cellular senescence holds a prominent role in cancer development, and studies have shown that senescent cells can limit tumor growth via tumor-suppressive genes such as p53 and induce cell cycle arrest (Huang et al., 2022). The p53-p21-Rb signaling pathway—an indirect transcriptional repression mechanism dependent on p53—downregulates multiple target genes associated with cell cycle progression, ultimately leading to cell cycle arrest (Engeland, 2022). Therefore, inducing cellular senescence within CRC tumors presents a promising therapeutic strategy for treating colorectal cancer.

Jianpi Huayu Decoction (JHD) is an established prescription formulated by Professor Ruiping Wang with nearly 30 years of clinical experience on the basis of Sijunzi Decoction (SJZD).SJZD is originated from the Chinese medical book “Prescriptions of the Bureau of Taiping People’s Welfare Pharmacy” in the Song Dynasty over 860 years ago, and has been widely employed as an adjunctive therapy for CRC due to its capacity to boost qi and eliminate dampness (Zhao et al., 2020; Shang et al., 2023). JHD is composed of eight herbs: *Pseudostellaria heterophylla* (Miq.) Pax (Tai Zi Shen, TZS); *Coix lacryma-jobi* L. (Yi Yi Ren, YYR); *Atractylodes macrocephala* Koidz. (Bai Zhu, BZ); *Astragalus mongholicus* Bunge (Huang Qi, HQ); *Salvia miltiorrhiza* var. *miltiorrhiza* (Dan Shen, DS); *Scutellaria barbata* D.Don (Ban Zhi Lian, BZL); *Paridis Rhizoma* (Chong Lou, CL), and *Curcuma zedoaria* (Christm.) Roscoe (EZ) (Table 1; plant names were obtained from http://www.worldfloraonline.org). JHD adds several herbs including DS, BZL, CL and EZ to SJZD to detoxify and dissolve blood stasis, specializing in the treatment of qi deficiency and blood stasis in CRC patients.

In our previous studies, we observed that JHD hindered the in vitro proliferation of CRC cells by provoking cell cycle arrest at the G0/G1 phase and promoting apoptosis (Xi et al., 2015). JHD also exhibited substantial inhibition of subcutaneous transplantation of LoVo colon cancer cells in nude mice (Ling et al., 2012). Our clinical research demonstrated that JHD combined with chemotherapy in advanced CRC could regulate the expression of miRNA-143 and miRNA-145, stabilize the tumor, improve the clinical symptoms, and enhance the quality of life without obvious toxic side effects (Xue et al., 2017). Nonetheless, the complete mechanism of action for JHD remains partially understood due to the intricate nature of its constituents and the challenge of investigating numerous targets within herbal formulations.

Recently, the network pharmacology has become a systematic method of drug discovery, revolutionizing the understanding and treatment of diseases (Nogales et al., 2022). It overcomes the limitations posed by the organ-centered and single-target therapy dogma, and enables precise and effective therapeutic interventions. Here, we investigated the underlying mechanisms by which JHD inhibits CRC development. We employed an integrated approach combining network pharmacology, bioinformatics and in vivo experiments to analyze the intricate interactions between JHD components and biological targets associated with CRC and cellular senescence. Using this integrated approach, we aimed to comprehensively understand the mechanisms underlying the inhibitory effects of JHD on CRC development.

Table 1
Herbs in Jianpi Huayu decoction.

Chinese Names	Herb Names in Latin	Abbreviations
Tai Zi Shen	*Pseudostellaria heterophylla (Miq.) Pax*	TZS
Yi Yi Ren	*Coix lacryma-jobi L.*	YYR
Bai Zhu	*Atractylodes macrocephala Koidz.*	BZ
Huang Qi	*Astragalus mongholicus Bunge*	HQ
Dan Shen	*Salvia miltiorrhiza* var. *miltiorrhiza*	DS
Ban Zhi Lian	*Scutellaria barbata D.Don*	BZL
Chong Lou	*Paridis Rhizoma*	CL
E Zhu	*Curcuma zedoaria (Christm.) Roscoe*	EZ

2. Materials and methods

2.1. Network-based pharmacological analysis of JHD

2.1.1. Identification of JHD bioactive compounds and their associated targets

The potentially active compounds in JHD were identified from the Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform database (TCMSP) (https://old.tcmsp-e.com/tcmsp.php) (Ru et al., 2014) as well as chromatography tandem mass spectrometry (HPLC-MS/MS) technology analysis (Liu et al., 2022). In TCMSP database, we employed pharmacodynamics to identify active ingredients that met two criteria: oral bioavailability (OB) $\geq$ 30 % and drug-likeness (DL) $\geq$ 0.18 (Zhang et al., 2020). The criterion of OB $\geq$ 30 % identifies compounds with good potential for oral absorption, making them suitable candidates for oral drug administration. A DL value of $\geq$0.18 is a common threshold for selecting compounds with characteristics conducive to drug development. These criteria serve as valuable initial filters for identifying compounds with favorable pharmacological and drug-like properties. HPLC-MS/MS is a powerful analytical technique that employs chromatography and mass spectrometry to identify and quantify small molecules in a sample. It provides detailed information about the sample’s composition, substance content, and nature, aiding in comprehensive sample characterization (Chen et al., 2013). Detailed methodological information for HPLC-MS/MS is available in the Supplementary file&Tables S1–S3. For further screening of the compounds, we considered absorption, distribution, metabolism, and excretion (ADME) properties as important indicators of the effectiveness of a drug candidate. The Canonical simplified molecular input line-entry system (SMILES) format of each compound was retrieved by PubChem (https://pubchem.ncbi.nlm.nih.gov/)(Kim et al., 2021) and submitted to SwissADME (http://www.swissadme.ch/)(Daina et al., 2017) to assess pharmacokinetics and druglikeness properties. We exclusively considered the compounds that exhibited high gastrointestinal absorption and strong drug similarity.

Target prediction for all active compounds of JHD was performed using the PubChem and SwissTargetPrediction database (http://www.swisstargetprediction.ch/) (Daina et al., 2019). Furthermore, we standardized the drug targets for each active ingredient using the UniProt database (https://www.uniprot.org/) (UniProt: the universal protein knowledgebase in 2021, 2021)—set for human species.

2.1.2. Collection of potential target genes for CRC

The targets associated with CRC were obtained by searching the keyword "colorectal cancer" in various databases, including OMIM (https://omim.org/search/advanced/) (Hamosh et al., 2021), GeneCards (https://www.genecards.org/) (Safran et al., 2010), Drugbank (https://go.drugbank.com/) (Wishart et al., 2018) and DisGeNET (https://www.disgenet.org/) (Pinero et al., 2020). Targets that scored above the median were selected as CRC disease targets from the GeneCards database, with duplicates removed.

2.1.3. Establishment of the compound-target (C-T) network

The intersecting target genes of JHD against CRC were obtained using the JVENN online software mapping tool platform (http://bioinfo.genotoul.fr/jvenn)(Bardou et al., 2014). The regulatory compound-target (C-T) network was mapped by the software Cytoscape version 3.9.1 (Otasek et al., 2019), in order to scientifically and rationally explain the relationship between compounds and targets.

2.1.4. GO and KEGG enrichment analyses

For the intersection targets of JHD against CRC, Gene Ontology (GO) functional annotation and Kyoto Encyclopedia of Genomes (KEGG) pathway enrichment analyses were performed based on the DAVID data platform (https://david.ncifcrf.gov/home.jsp) (Sherman et al., 2022). We selected "human species" on this platform and then analyzed the biological processes (BP), cellular components (CC), molecular functions (MF), and KEGG signaling pathways related to CRC and JHD. The enrichment analyses were all filtered with a false discovery rate of less than 0.01 as well as a minimum count of three. Using the R software (version 4.3.1), the top 10 GO terms and the top 20 KEGG pathways were visualized according to their count values. Notably, KEGG enrichment analysis highlighted cellular senescence as one of the primary signaling pathways associated with JHD in CRC treatment.

2.1.5. Screening of hub targets for JHD-induced cellular senescence in CRC

We obtained cellular senescence-related genes from the GSEA (https://www.gsea-msigdb.org/) database (Mootha et al., 2003; Subramanian et al., 2005) and overlapped these targets with JHD- and CRC-related targets using the JVENN online software mapping tool platform (Bardou et al., 2014). By identifying intersecting targets, we aimed to detect the common genes associated with cellular senescence and targeted by both JHD and CRC. Protein-protein interactions (PPI) play a crucial role in the intricate network of functional associations between biological macromolecules in cellular life. To explore the interactions between hub genes, we used the STRING database (version 11.5) (https://string-db.org/cgi/input) (Szklarczyk et al., 2021). The PPI network was constructed by importing hub genes with species limited to "Homo sapiens". To ensure a clear visualization of the PPI network, we set a minimum required interaction score of medium confidence (0.400) and excluded unconnected nodes. The resulting network was visualized using Cytoscape 3.9.1 software, enabling a clear representation of the intersecting targets. Subsequently, we identified 10 targets with the highest node degree values as hub targets for molecular docking and further bioinformatics analysis.

2.1.6. Construction of the target-pathway (T-P) network

The top 20 signaling pathways with the highest count values obtained from the KEGG enrichment analysis were imported into Cytoscape software. The corresponding targets for each pathway were mapped to construct the main target-pathway (T-P) network, providing insights into the pivotal targets and linked pathways implicated in JHD's treatment of CRC.

2.1.7. Molecular docking

The two-dimensional structures of the potentially active compounds in JHD were retrieved from the TCMSP database in MOL2 format. Meanwhile, the X-ray crystal structures of the hub targets were obtained from the Worldwide Protein Data Bank (PDB) database (https://www.rcsb.org/) (Goodsell et al., 2020), specifically selecting structures from "Homo sapiens" species with resolutions ranging from 2.0 to 3.0° A. To assess the binding affinity between the compounds and targets, we utilized Autodock Vina 1.1.2 (Eberhardt et al., 2021; Trott and Olson, 2010) for molecular docking. AutoDock Vina has shown a significant improvement in the average accuracy of combined mode prediction compared to AutoDock 4 (Trott and Olson, 2010). PyMOL 4.6.0 (Baugh et al., 2011) was used to display the docking results of the hub targets with the main active compounds. A binding energy threshold condition $\leq$5.0 (BE, kcal/mol) was considered the most favorable docking pose.

2.2. Bioinformatics analysis

2.2.1. Expression analysis of hub targets

We used the Gene Expression Profiling Interactive Analysis 2 (GEPIA2) online tool (http://gepia2.cancer-pku.cn/#index) (Tang et al., 2019) to examine the gene expression levels of the target genes in normal and CRC tissues. In GEPIA2, we selected our hub targets as the gene set of interest and chose "COAD" and "READ" as the two cancer types for comparison, using data from The Cancer Genome Atlas (TCGA) project. We downloaded the transcriptome and clinical data of CC from the GEO (https://www.ncbi.nlm.nih.gov/geo/) (Barrett et al., 2013) database. Specifically, we used the GSE9348, GSE32323, GSE74602, and GSE113513 datasets for our analysis. Using the limmaR package (version 3.54.2) in R software (v 4.3.1), we validated the expression of hub targets in these datasets. The final gene expression levels were determined by calculating the mean values of various probes in the GEO dataset. When the value was $\leq$ zero, a log2 transformation was performed. Detailed information on the GSE datasets used in this analysis is presented in Table S4.

2.2.2. Survival analysis of cellular senescence-related targets

TP53, CDKN1A, RB1, and MDM2 are key targets and regulators of the p53-p21-Rb signaling pathway, which is closely linked to cellular senescence. Conversely, IL6 is a member of the senescence-associated secretory phenotype (SASP) and is associated with the pro-inflammatory state observed in senescent cells. Here, we keyed TP53, CDKN1A, RB1, MDM2, and IL6 into GEPIA2 to investigate their impact on overall survival in TCGA-COAD and TCGA-READ datasets. This can provide insight into their potential prognostic value and shed light on their role in disease progression and patient outcomes.

2.3. Experimental validation in vivo

2.3.1. Animals

Human colon cancer LoVo cells were injected subcutaneously into the right flank of athymic BALB/c nude mice to establish the cancer xenograft model (4 weeks old, 18–22 g) (2×10^6/mouse) (Nanjing University of Chinese Medicine, Nanjing, China). The animals were housed in a specific pathogen-free facility at the Animal Center of Nanjing University of Traditional Chinese Medicine (Nanjing, China), where they had ad libitum access to food and water. The environment was controlled with a temperature of 22 ± 2 °C and humidity maintained at $50 \pm 10\%$. The research adhered to the principles of the Basel Declaration and the guidelines outlined by the National Institutes of Health (Maryland, USA). Approval for all animal experiments was obtained from the Nanjing University of Chinese Medicine Animal Care and Use Committee, and the experiments were conducted in accordance with NIH guidelines.

2.3.2. Quality control of JHD and drug treatments

JHD is a compound preparation of Chinese herbal medicine, all of whose ingredients are procured from the Jiangsu Provincial Hospital of Traditional Chinese Medicine and decocted in the central laboratory according to the concentration requirements. All herbal ingredients were sourced from reputable suppliers who adhere to Good Manufacturing Practices (GMP). The herbs used were taxonomically identified and authenticated to ensure they met the required standards and were free from adulteration. We followed established quality control standards for herbal materials, which include criteria for appearance, odor, taste, and chemical composition. These standards are in accordance with the Chinese Pharmacopoeia and other relevant regulatory guidelines. The preparation of JHD and its components was carried out in a laboratory that adheres to Good Laboratory Practices (GLP). This ensures the proper handling, storage, and preparation of herbal materials. Briefly, *Pseudostellaria heterophylla* (Miq.) Pax (15g), *Astragalus mongholicus* Bunge (15g), *Atractylodes macrocephala* Koidz. (15g), *Coix lacryma-jobi* L. (30g), *Paridis Rhizoma* (15g), *Curcuma zedoaria* (Christm.) Roscoe (15g), *Salvia miltiorrhiza* var. *miltiorrhiza* (15g), *Scutellaria barbata* D.Don (15g) were heated at 100 °C to make three different concentrations of aqueous decoctions. Detailed records were maintained throughout the herbal procurement and preparation process, including information on the source and batch numbers.

JHD was administered daily by intraperitoneal injection, starting on day 5 after inoculation with cancer cells for 14 days. The mice were randomly categorized into five groups: control (CON), JHD low-dose (JHD-L), JHD medium-dose (JHD-M), JHD high-dose (JHD-H), and 5-FU (5-FU: a clinical CRC-positive drug), with five mice in each group. To make it equivalent to the oral human dosage, the JHD-L, JHD-M and

JHD-H groups were treated orally once a day with JHD at 0.88, 1.75, and 3.50 g/ml, respectively. The 5-FU group was orally administered 25 mg/kg 5-FU once a day. At the end of the experiments, segments of each tumor from the diverse groups were gathered, frozen, and subjected to paraffin-sectioning for immunohistochemistry and senescence-associated β-galactosidase (SA-β-Gal) staining. The remaining tumors from the various groups were gathered and lysed to perform qRT-PCR and western blotting analyses. The tumors were photographed and weighed, and all data were recorded.

2.3.3. Immunohistochemistry

Immunohistochemical staining was conducted following the protocol outlined by Dunstan et al. (2011). Tissue samples, cut to a thickness of 4 μm, were prepared and affixed onto slides. Staining was executed using polyclonal antibodies targeting the following biomarkers of cellular senescence: lamin b1 (catalog no. 12987-1-AP; Wuhan Sanying Biotechnology, 1:100), IL-6 (catalog no. 21865-1-AP; Wuhan Sanying Biotechnology), and $p16^{Ink4a}$ (catalog no. 10883-1-AP; Wuhan Sanying Biotechnology, 1:100). Photomicrography was captured using an IX-51 microscope.

2.3.4. Senescence-associated β-galactosidase (SA-β-Gal) staining

SA-β-Gal activity at pH 6.0 was cytochemically detected using cytochemical methods as per the manufacturer's instructions (Beyotime, Shanghai, China). Briefly, we fixed and stained tumor cells using the solution provided in the kit. The stained cells were examined and captured under a microscope.

2.3.5. Quantitative real-time polymerase chain reaction (qRT-PCR)

Based on the findings of network pharmacology and bioinformatics analyses, we opted to investigate P53 (TP53), p21 (CDKN1A), RB1, and IL-6 as the candidates for elucidating the mechanism through which JHD induces cellular senescence in CRC. Additionally, the mRNA expression of P16 was also predicted to be another cell cycle regulator. Total RNA was isolated from cultured cells using TRIzol reagent, followed by phenol-chloroform phase separation as per the manufacturer's guidelines. Subsequently, reverse transcription was carried out utilizing the HiScript II Q RT SuperMix Kit (Vazyme, Nanjing, China). We used the ChamQTM Universal SYBR QPCR Master Mix for qPCR analysis. The primer sequences used for qRT-PCR are enumerated in Table 2.

2.3.6. Western blot

Western blotting was employed to detect the protein expression levels of P53, P21, RB1, P16, and IL-6. Protein concentrations were measured using a BCA protein assay kit after extracting total protein with RIPA buffer. The protein samples were transferred to PVDF membranes and then incubated at 4 °C overnight with primary antibodies against P53 (dilution 1:2000), P21 (dilution 1:2000), RB1 (dilution 1:3000), P16 (dilution 1:2000), IL-6 (dilution 1:1000), and β-actin (dilution 1:3000). Subsequently, HRP-conjugated secondary antibodies were applied. Grayscale values were quantified using Image J software.

2.3.7. Statistical analysis

GraphPad Prism software was employed for the statistical analysis of the experimental results. One-way analysis of variance (ANOVA) was utilized to compare the variations in the experimental data across the groups. A significance level of $P < 0.05$ was considered statistically significant.

Table 2
Real-Time polymerase chain reaction primers.

Gene	Forward Primer (5′-3′)	Reverse Primer (5′-3′)
P53	CCTCACCATCATCACACTGG	TCTTGCGGAGATTCTCTTCC
RB1	AGGTCTGCCAACACCAACAA	CTGCTTTTGCATTCGTGTTCG
P16	TTCCTGGACACGCTGGTGGT	CTATGCGGGCATGGTTACTGC
IL-6	GGAGACTTGCCTGGTGAA	AAAGCTGCGCAGAATGAGAT
p21	CACCACTGGAGGGTGACTTC	ATCTGTCATGCTGGTCTGCC

3. Results

3.1. Network pharmacology analysis

3.1.1. The interaction network between JHD and CRC targets

Firstly, we obtained 376 compounds from the TCMSP database and HPLC-MS/MS analysis (Fig. 1A) in total. Then, 129 potential active ingredients were screened from the components of JHD according to the ADME (Absorption, Distribution, Metabolism and Excretion) criteria available on the SwissADME website. Total ion flow (TIC) plots of JHD scanned by HPLC-MS/MS were shown in Fig. 1B. The results of the HPLC-MS/MS identification are detailed in the Supplementary file&Tables S5–S8. Prototype ingredients and sources of JHD were presented in Table S9.

Secondly, after predicting the database and removing duplicate values, we obtained 678 potential targets for the active compounds and 1305 targets for CRC. By merging the two target sets, we identified 214 common targets (Fig. 1C). Meanwhile, a "JHD C-T″ network was constructed (Fig. 1D), in which quercetin, isoflavone, bisdemethoxycurcumin, (3R)-3-(2-hydroxy-3,4-dimethoxyphenyl)chromium-7-ol, 12-selenosyl-2E,8E,10E-atractylenetriol, microstegiol, and luteolin are considered to be the main active components of JHD-targeted therapy for CRC.

3.1.2. Enrichment analyses

To gain insight into the potential intracellular processes influenced by JHD in CRC, we conducted GO and KEGG pathway enrichment analyses of intersecting targets between JHD and CRC. The foremost 10 pertinent functions were extracted from the GO-CC, GO-MF, and GO-BP functional annotations based on the 214 genes involved (Fig. 2A). As shown in Fig. 2B, the KEGG enrichment analysis unveiled numerous notable pathways, including infection, lipid and atherosclerosis, hepatitis, microRNAs in cancer, MAPK signaling pathways, chemical carcinogenesis-receptor activation, Ras signaling pathways, measles, cellular senescence, PI3K-Akt signaling pathways, and AGE-RAGE signaling pathways in diabetic complications. This suggests that cellular senescence is one of the primary potential pathways for the JHD-targeted treatment of CRC. Further investigations into the role of cellular senescence in JHD treatment may yield valuable insights into the underlying mechanisms and potential therapeutic strategies for addressing CRC.

3.1.3. Screening of hub targets for JHD-induced cellular senescence in CRC

In total, 198 cellular senescence-associated targets were obtained from the GSEA database (Human Gene Set: REACTOME_CELLULAR_SENESCENCE). The Venn diagram indicated the presence of 21 shared genes, obtained by intersecting genes associated with active ingredients, CRC, and cellular senescence (Fig. 3A). Based on the degree values, STAT3, TP53, JUN, CDKN1A, MDM2, MAPK3, FOS, RB1, IL6, and CDK4 were identified as ten hub genes for JHD-induced cellular senescence in CRC. Concurrently, a PPI network diagram of the hub targets was formulated (Fig. 3B–C). Additionally, we visualized the leading 20 signaling pathways with the highest count values alongside their respective targets, resulting in a "T-P″ network featuring 183 nodes and 935 edges (Fig. 3D). Notably, cellular senescence was among the most significantly affected pathways, confirming the relevance of JHD in treating CRC-induced cellular senescence.

3.1.4. Molecular docking

To better understand the signaling pathways and interactions among active components and hub targets, we performed molecular docking of ingredients, including quercetin, isoflavanone, bisdemethoxycurcumin, (3R)-3-(2-hydroxy-3,4-dimethoxyphenyl)chroman-7-ol, 12-senecioyl-

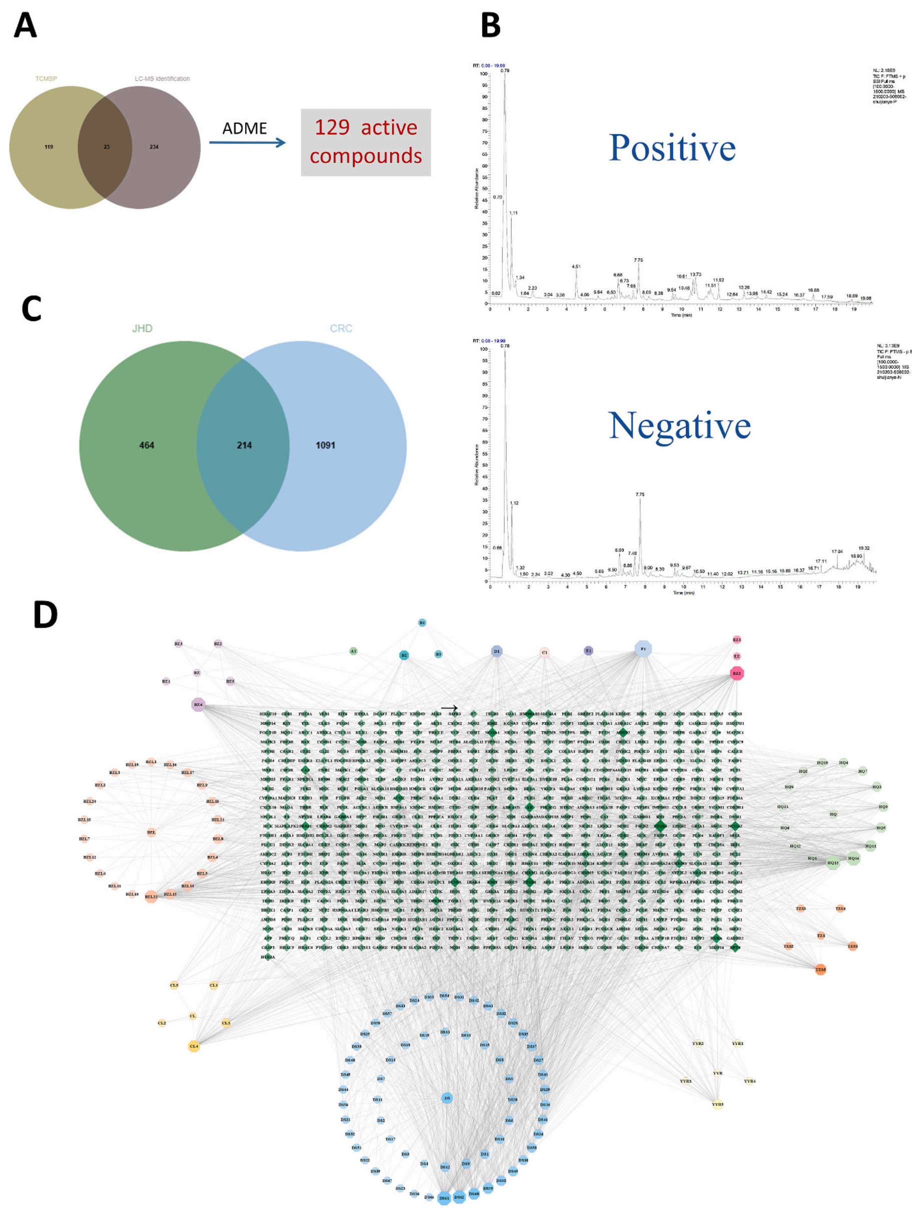

Fig. 1. The interaction network between JHD targets and CRC targets. (A)Compounds in JHD obtained by LC-MS identification, TCMSP database and ADME screening. (B) Total ion flow (TIC) plots of JHD in positive- and negative-ion modes. (C)The overlapped targets between the JHD and CRC. (D) The Compound-Target (C–T) network. The green rhombic means target; three circles, four pentagons and a pink vertical line mean constituent of JHD. Besides, the other's letters mean repetitive ingredients. A1, B1, B2, B3, C1, D1, E1 and F1 represents (3S,8S,9S,10R,13R,14S,17R)-10,13-dimethyl-17-[(2R,5S)-5-propan-2-yloctan-2-yl]-2,3,4,7,8,9,11,12,14,15,16,17-dodecahydro-1H-cyclopenta [a]phenanthren-3-ol, CLR, stigmasterol, sitosterol, beta-sitosterol, luteolin, hederagenin, and quercetin respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2E,8E,10E-atractylentriol, microstegiol, and luteolin, plus targets including JUN, STAT3, TP53, CDKN1A, FOS, MAPK3, MDM2, IL6, RB1, and CDK4. The molecular docking results are depicted in the form of a heatmap (Fig. 4A). As illustrated in Table 3, The binding affinities of the compound to the crystal structure of the protein were below −5 kcal/mol, suggesting favorable binding interactions. The molecular docking results of TP53, CDKN1A, IL6, and RB1 with their respective main active compounds were visualized using PyMol software (Fig. 4B).

3.2. Bioinformatics analysis

3.2.1. Expression of hub targets in colorectal tissue of patients with CRC

The results obtained from the GEPIA2 and GEO databases demonstrated significant differences in the expression levels of hub genes between patients with CRC and controls. Specifically, CDK4, IL6, RB1, and TP53 exhibited increased expression, whereas CDKN1A, MAPK3, MDM2, and STAT3 were downregulated (Fig. 5(A–B)). This suggests that the expression of our hub targets—especially MDM2, TP53, CDKN1A, RB1, and IL6, which are closely related to cellular senescence—may be associated with CRC development.

3.2.2. Survival analysis of cellular senescence-related targets

Survival analysis of cellular senescence-related targets (Fig. 6) indicated a notably distinct prognostic significance for CDKN1A ($P = 0.045$). The Kaplan–Meier curve demonstrated that individuals with elevated CDKN1A expression exhibited longer survival, suggesting that CDKN1A—a gene closely associated with cellular senescence—may serve as a favorable prognostic marker for CRC.

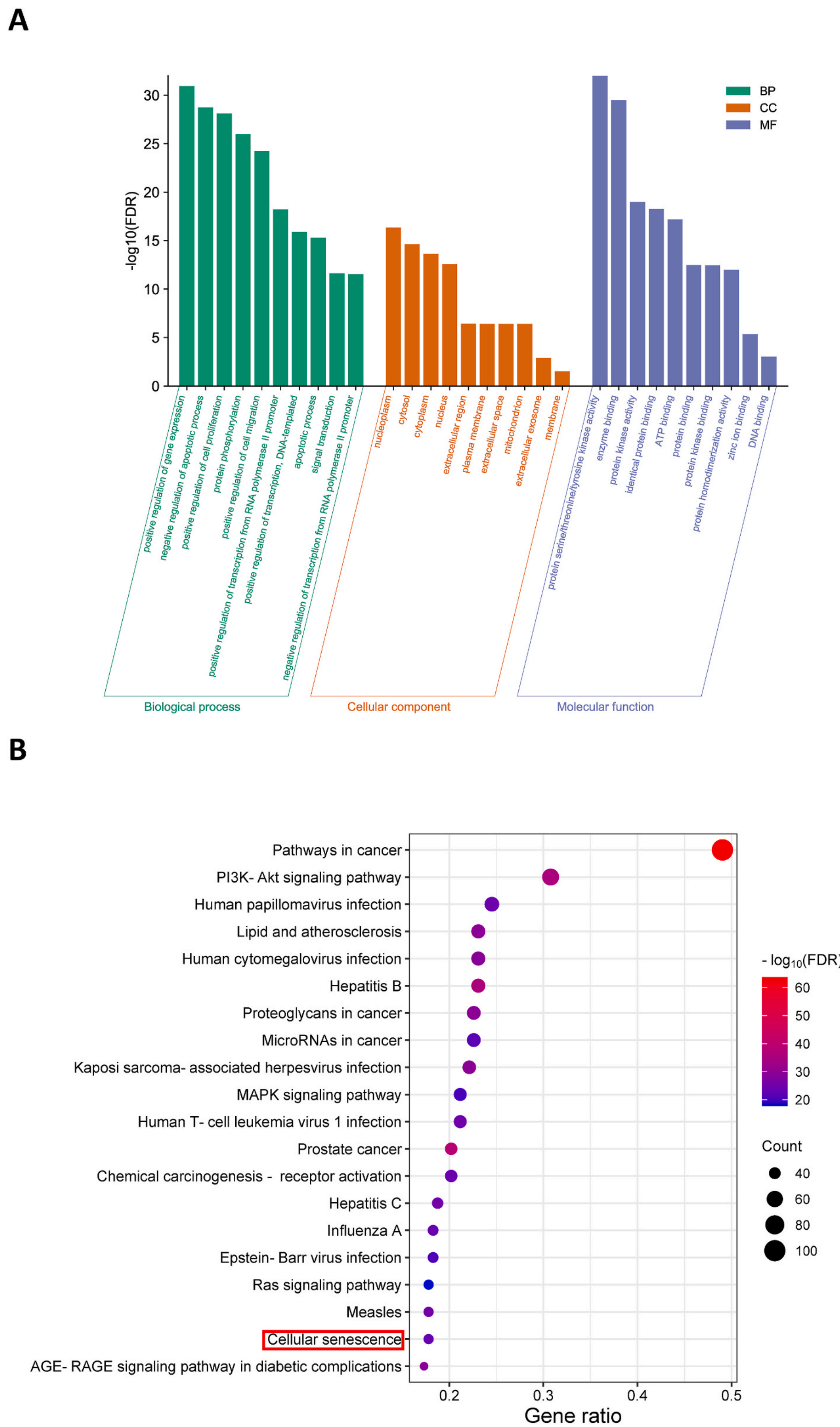

Fig. 2. Enrichment analysis for the overlapped targets between the JHD targets and CRC targets. (A) The top 10 relevant functions were from GO-BP, GO-CC, GO-MF ($p \leq 0.01$). (B) The 15 most enriched KEGG pathways, including Cellular senescence ($p \leq 0.01$).

3.3. Experimental validation in vivo

3.3.1. JHD inhibits tumor growth in vivo

Next, we conducted in vivo experiments using a xenograft mouse model with human colon cancer LoVo cells to corroborate our network pharmacology findings and results. Following one week of cancer cell inoculation, intraperitoneal injections of JHD and 5-FU were administered over a span of 14 days. Subsequently, the tumors were gathered to assess cell proliferation, signaling activation, and induction of senescence. We noted a gradual growth of LoVo tumor cells in the mice. Nevertheless, administration of JHD substantially curbed tumor progression while not influencing weight

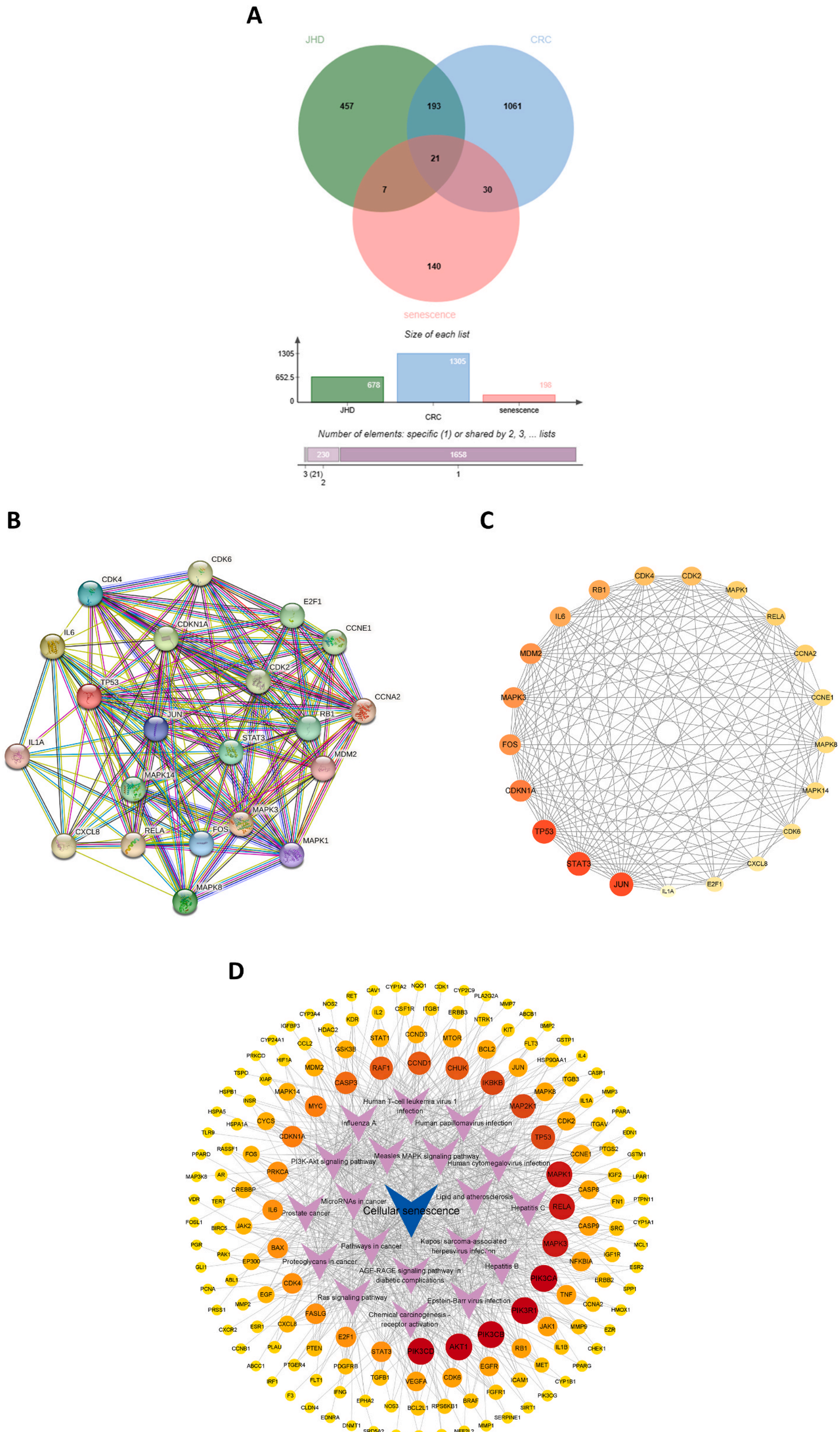

Fig. 3. Construction of PPI network and screening of hub targets for JHD induced cellular senescence in CRC. (A) The overlapping genes of JHD, CRC and cellular senescence. (B–C) The PPI network of JHD, CRC and cellular senescence. The darker the red color, the greater the degree value. The lighter the yellow color, the smaller the degree value. (D) The Target-Pathway (T–P) network. Cellular senescence is among the top 20 signaling pathways with the highest count values. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

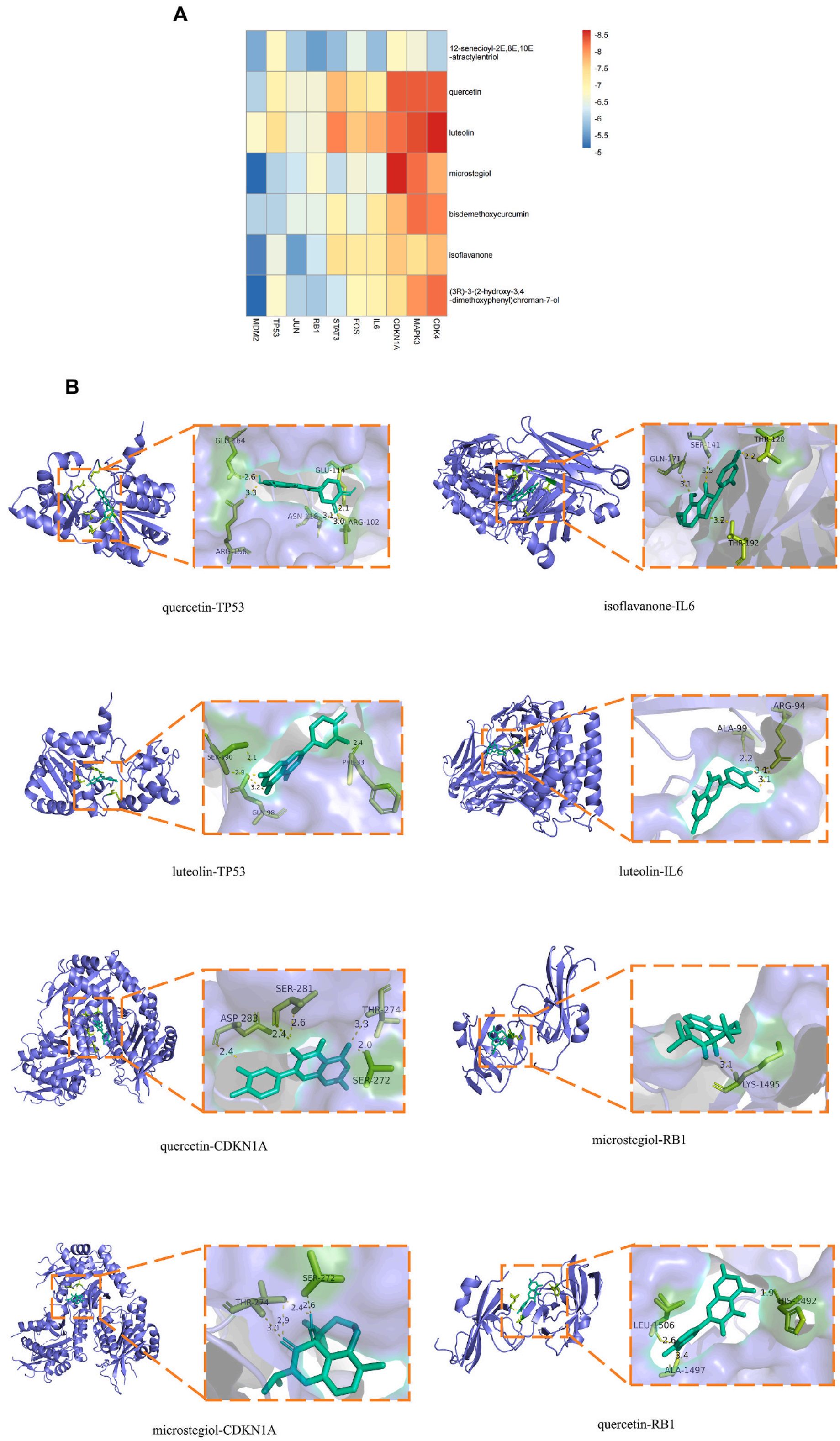

Fig. 4. Molecular docking analysis for the core hub targets. (A) Heatmap of binding between main active components and hub targets. The darker the red color, the better the binding activity. (B)) The 3D molecular docking structures of key compounds and key target genes. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

gain (Fig. 7(A-C)). Among the three different JHD administration doses, the JHD-H group showed inhibited tumor growth, second only to the 5-FU group.

3.3.2. JHD induced senescence and activated p53-p21-Rb pathways in colon cancer cells in vivo

Representative immunohistochemical images of all tissues analyzed for p16^{Ink4a}, IL-6, and lamin b1 expression are shown in Fig. 8A. In

Table 3
The docking scores of key constituents and hub targets.

Target	PDB ID	quercetin (kcal/mol)	isoflavanone (kcal/mol)	bisdemethoxycurcumin (kcal/mol)	(3R)-3-(2-hydroxy-3,4-dimethoxyphenyl)chroman-7-ol (kcal/mol)	12-senecioyl-2E,8E,10E-atractylentriol (kcal/mol)	microstegiol (kcal/mol)	luteolin (kcal/mol)
JUN	5fv8	−6.6	−5.5	−6.4	−6.0	−5.9	−6.2	−6.5
STAT3	5ax3	−7.7	−7.4	−7.0	−6.2	−5.8	−6.1	−8.1
TP53	1yc5	−7.0	−6.5	−6.0	−6.7	−6.8	−6.0	−7.4
CDKN1A	7kys	−8.3	−7.6	−7.7	−7.4	−6.8	−8.6	−8.2
FOS	2c1n	−7.4	−7.2	−6.4	−6.9	−6.4	−6.6	−7.6
MAPK3	4qtb	−8.3	−7.4	−8.2	−8.0	−6.6	−8.2	−8.4
MDM2	3jzr	−6.0	−5.3	−6.0	−5.1	−5.6	−5.1	−6.7
IL6	4zs7	−7.1	−7.2	−7.2	−7.0	−5.8	−6.4	−7.8
RB1	7ea2	−6.6	−6.2	−6.4	−5.9	−5.5	−6.7	−6.6
CDK4	2w99	−8.3	−7.7	−8.1	−8.2	−6.0	−7.8	−8.6

contrast to the negative control group, the expression levels of $p16^{Ink4a}$ were markedly elevated in the JHD-L, JHD-H, and 5-FU groups, whereas those of IL-6 and Lamin b1 were reduced to different degrees in the JHD-M, JHD-H, and 5-FU groups. Notably, JHD treatment induced a notable elevation in the activity of SA-β-Gal, a senescence marker. The results showed more pronounced SA-β-Gal staining in the JHD-M, JHD-H, and 5-FU groups when contrasted with the negative control group, especially notable in the JHD-H and 5-FU groups.

We used qRT-PCR to assess the impact of JHD on the mRNA expression of TP53, P21, RB1, P16, and IL-6 (Fig. 8B). The findings revealed that the mRNA expression levels of TP53, P21, and RB1 were notably higher in the three distinct JHD groups compared to the control group ($P < 0.05$), yet lower compared to the positive control (5-FU) group. Among the three JHD groups, JHD-H exhibited the strongest effects. For $p16^{Ink4a}$, its expression demonstrated a significant increase in both the JHD-L and JHD-H groups ($P < 0.05$), while the JHD-M group exhibited no statistically significant effect. IL-6 expression showed a noticeable decrease in the JHD-M and JHD-H treatment groups, indicating a potential inhibitory impact of JHD on SASP secretion.

Western blotting results indicated a noteworthy increase in protein levels of TP53, P21, and RB1 across all three JHD treatment groups, yet the protein level of P16 increased solely in the JHD-H group ($P < 0.05$) (Fig. 8C). In addition, IL-6 protein levels were reduced in the JHD-H group, whereas no significant alterations were noted in the remaining two JHD treatment groups. Together, these results suggest that JHD has a positive effect on the p53-p21-Rb cellular senescence-associated pathway as well as on the cell cycle inhibitor $p16^{Ink4a}$, while it exhibited an inhibitory effect on the secretion of IL-6, an SASP. The strongest therapeutic effect was observed in JHD-H group.

4. Discussion

Colorectal cancer (CRC) ranks as the third most prevalent cancer globally and stands as the second leading contributor to cancer-related deaths. This malignancy is characterized by its propensity for metastasis and an unfavorable prognosis. The medical predicament surrounding CRC involves limited treatment options with potential side effects, compounded by drug resistance challenges and suboptimal outcomes experienced by a subset of patients (Dekker et al., 2019). In comparison, TCM treatment demonstrates distinct advantages in anti-tumor therapy, including its multi-targeting nature, reduced side effects, and improved efficacy (Wang et al., 2021). During the clinical treatment process, we observed the therapeutic effect of JHD in patients with CRC. However, the precise underlying mechanism remains unclear and requires further investigation. We conducted further research to explore whether JHD induces senescence in CRC cells and whether the p53-p21-Rb signaling pathway serves as one of the mechanisms through which JHD triggers cellular senescence in CRC.

Here, we identified cellular senescence as one of the mechanisms through which JHD exerts its anti-CRC effects based on network pharmacology and enrichment analysis. The key active compounds of JHD include quercetin, isoflavones, bisdemethoxycurcumin, (3R)-3-(2-hydroxy-3,4-dimethoxyphenyl)chromium-7-ol, 12-selenosyl-2E,8E,10E -atractylenetriol, microstegiol, and luteolin. These components have been observed to exert antitumor activities by controlling tumor cell apoptosis (Maugeri et al., 2023), increasing ER-β expression in tumor tissues (Yu et al., 2016), enhancing immune function (Jung et al., 2016) and promoting oxidative transformation (Gordon et al., 2015). This indicates that the anti-CRC effects of JHD are likely driven by interactions between active compounds and their respective targets. Leveraging the common target genes of JHD in CRC, we conducted GO-BP, GO-CC, GO-MF, and KEGG enrichment analyses. The results of KEGG enrichment analysis highlighted that the activation of cellular senescence stands as a key pathway involved in JHD's therapeutic effects on CRC. We therefore investigated and validated the molecular mechanism through which JHD treats CRC by inducing cancer cell senescence. As we established a "T-P″ network integrating targets from the intersection of JHD, CRC, and cellular senescence-related genes with the leading 20 signaling pathways from KEGG enrichment analysis, it became evident that the cellular senescence pathway maintained a significant connection to the intersected targets.

Next, we identified hub targets for JHD-induced cellular senescence in CRC through analysis of the PPI network. Here, the recognized hub targets encompassed STAT3, TP53, JUN, CDKN1A, MDM2, MAPK3, FOS, RB1, IL6, and CDK4. Among these targets, MDM2, TP53, CDKN1A, RB1, and CDK4 are key genes in the cellular senescence-associated signaling pathway p53-p21-Rb. MDM2, or mouse double minute 2 homolog, functions as an E3 ubiquitin ligase that negatively regulates TP53 by promoting its degradation (Konopleva et al., 2020). TP53, also known as tumor protein p53, serves as a tumor suppressor gene responsible for governing the expression of diverse genes associated with cell cycle arrest, DNA repair, and apoptosis (Donehower et al., 2019), promoting CRC development and progression. CDKN1A, also known as p21, is a cyclin-dependent kinase inhibitor activated by TP53 which functions as a regulator of senescence-induced cell cycle progression (Brenner et al., 2020). Aberrant CDKN1A expression was associated with CRC development and prognosis. RB1, or retinoblastoma 1, is a tumor suppressor gene that inhibits cell cycle progression by interacting with a range of proteins engaged in regulating cell cycle control (Knudsen et al., 2019). The p53-p21-Rb signaling pathway, linked with cellular senescence, is modulated by p53. In response to cellular stress, p53 triggers the expression of p21 (CDKN1A), which subsequently causes cell cycle arrest and curbs RB phosphorylation. This orchestration regulates cellular growth and apoptosis (Engeland, 2022). CDK4 interacts with RB and promotes its phosphorylation, releasing E2F and promoting cell cycle progression (Knudsen et al., 2019). Additionally, IL6, a SASP, contributes to the pro-inflammatory state of senescent cells, which can create a microenvironment conducive to tumor development and progression (Birch and Gil, 2020).

To verify whether these genes are therapeutic targets for JHD to

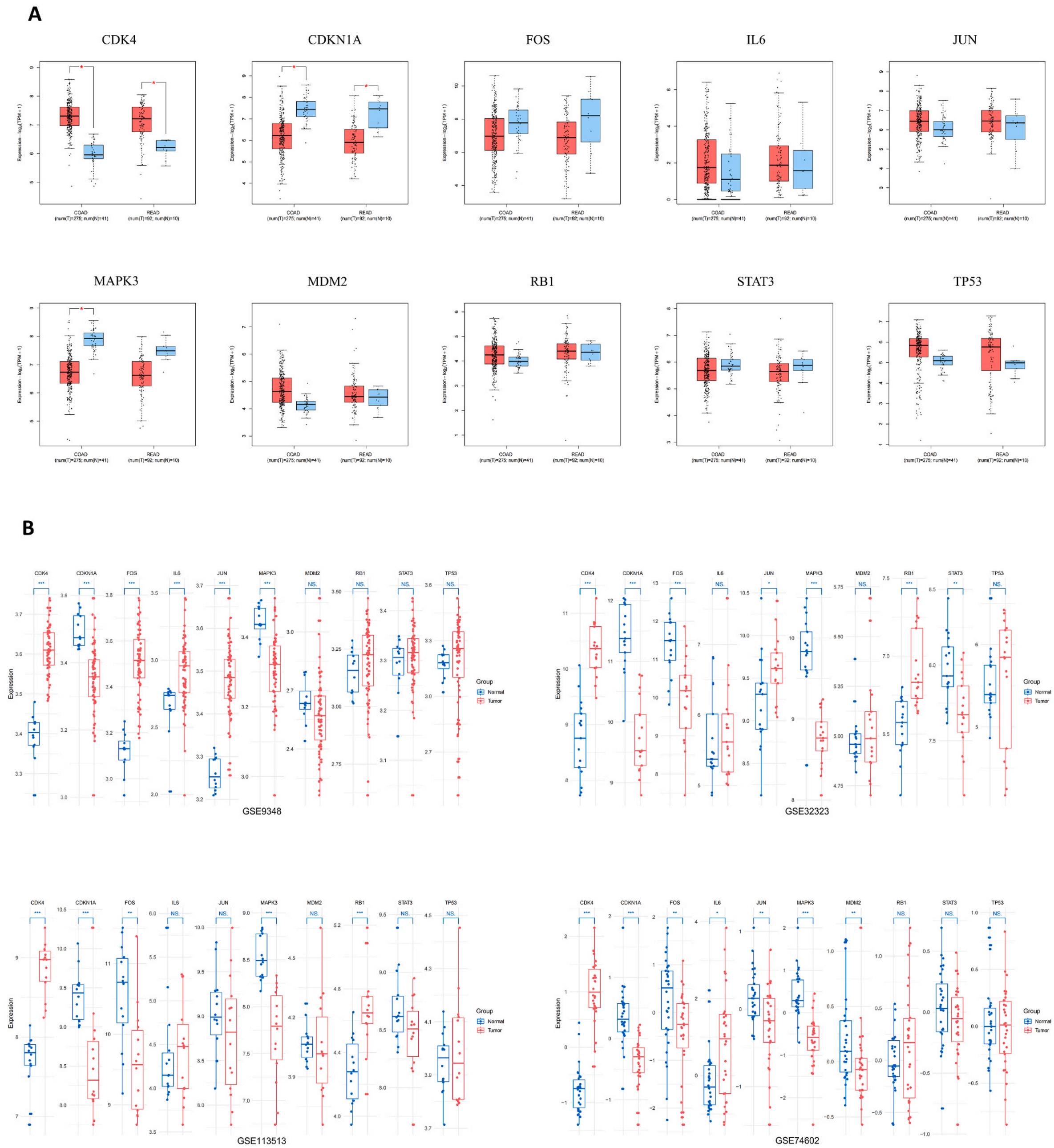

Fig. 5. Expression of hub targets in colorectal tissues of CRC patients. (A) Box plots showing the mRNA expression levels of hub targets in the GEPIA2 database. Red represents Tumor, blue represents normal. (B) Box plots showing the expression levels of hub targets in the GEO database. Red represents Tumor, blue represents normal. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

inhibit CRC by inducing cellular senescence, we performed molecular docking, bioinformatics analysis, and in vivo experiments in mice.

Our molecular docking results revealed that the seven key active ingredients of JHD exhibited favorable binding activity with hub targets. The targets—TP53, CDKN1A, RB1, and IL6, which are closely related to cellular senescence—showed strong binding energies for the active compounds. Among the docking network, the microstegiol-CDKN1A was the highest docking network (binding energy −8.6 kcal/mol). Microstegiol, a well-known diterpenoid isolated from *Salvia lachnocalyx (S. lachnocalyx)*, possesses cytotoxicity and promising anticancer potential (Hadavand Mirzaei et al., 2020), though there is still not enough evidence to prove the relationship between microstegiol and CRC. Quercetin has a significant inhibitory effect on CRC, and its mechanisms include cell growth reduction, cell cycle arrest, down-regulation of cell cycle-related factor mRNA, induction of cell apoptosis (Maugeri et al., 2023; Tang et al., 2020), which are closely related to cellular senescence. The powerful antioxidant and anti-inflammatory effects of luteolin play a great role in the treatment of colon cancer and related complications, mainly manifested as the inhibition of iNOS, COX-2, MMP-2 and MMP-9 expression, thereby inhibiting tumor angiogenesis (Imran et al., 2019). Isoflavones are the main components of soybean. A meta-analysis reported an inverse association between soy isoflavone intake and CRC risk (RR = 0.77, 95% CI: 0.72–0.82, $P = 0.024$)(Yu et al., 2016), possibly through dose-dependent G2/M cell cycle arrest and down-regulation of cyclin A expression (Bielecki et al., 2011). It has been shown that the curcumin derivative, bisdemethoxycurcumin, is an antiproliferative agent that inhibits cell growth by simultaneously inhibiting the G1/S and G2/M cycles and induces irreversible double-strand breaks in cancer cells by targeting topoisomerase IIα activity and expression (Belluti et al., 2013). At present, the effects of

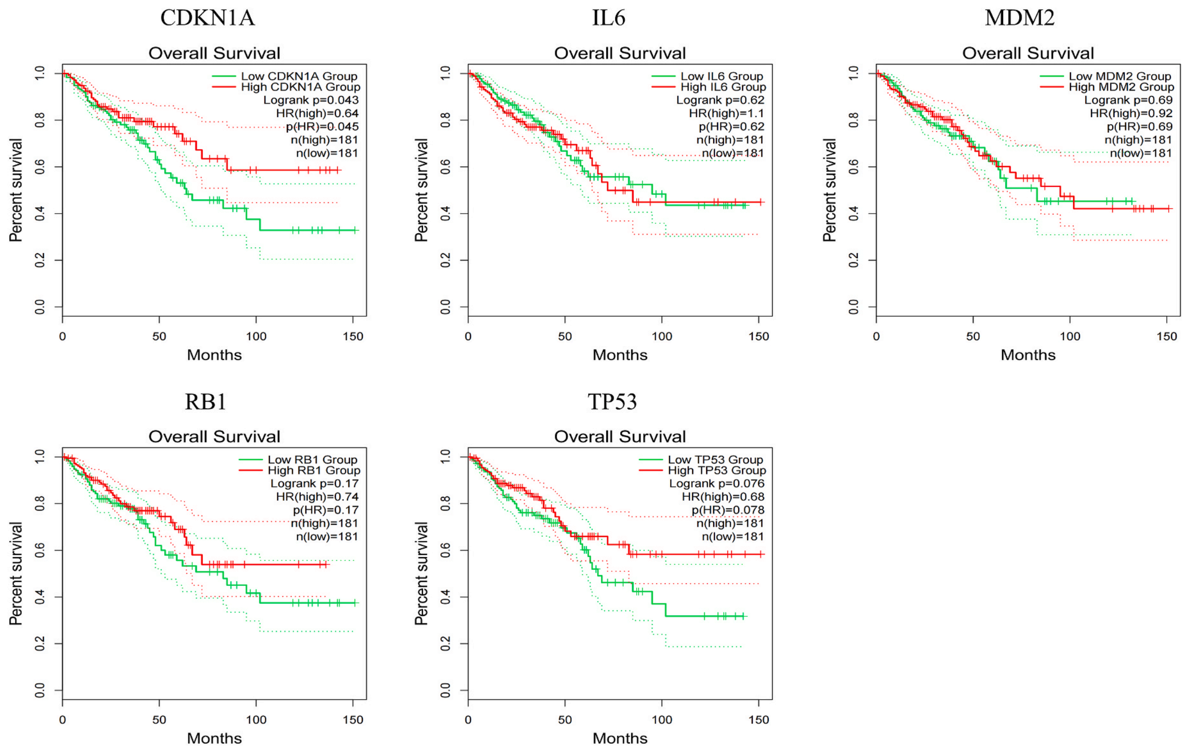

Fig. 6. Survival analysis of cellular senescence-related targets. The survival curve comparing the patients with high (red) and low (green) expression in CRC. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

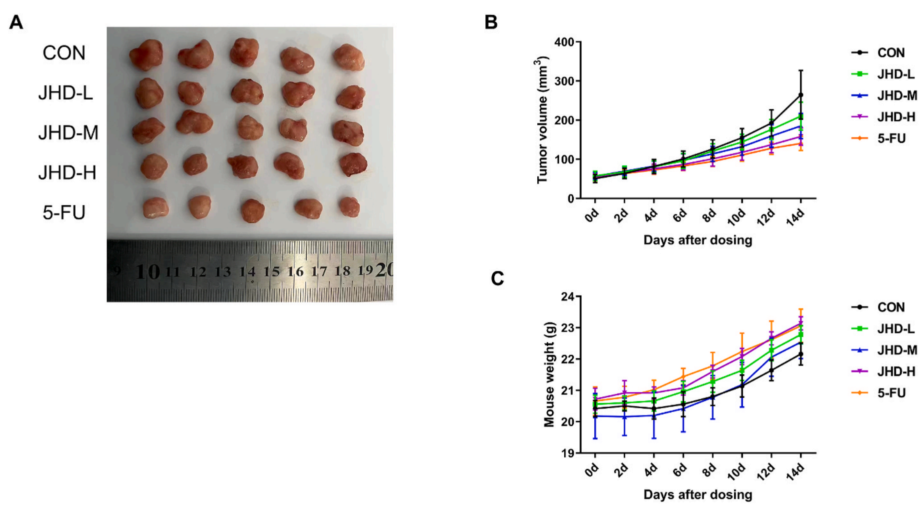

Fig. 7. Efficacy of drugs on tumor growth and body weight in the xenograft mouse model. (A) Photographs of control and treated tumors. (B–C) Tumor volume and mouse weight time course in nude mice. Tumor volume and mouse weight were measured at the indicated time points. Data are means ± SEM (n = 5).

(3R)-3-(2-hydroxy-3,4-dimethoxyphenyl)chroman-7-ol and 12-senecioyl-2E,8E,10E-atractylentriol on CRC have not been reported. In the future, more studies are needed to investigate whether and how these two compounds affect cellular senescence in CRC.

We identified the hub targets using the GEPIA2 and GEO databases, which demonstrated that IL6, RB1, TP53, and CDKN1A mRNA levels were notably elevated in CRC tissues. Additionally, the prognostic value of CDKN1A exhibited significant divergence ($P < 0.05$). The findings from the aforementioned analyses largely aligned with prior research outcomes.

To gain deeper insight into the interplay between JHD and senescence in CRC cells, we selected the p53-p21-Rb pathway and related targets for in vivo experimental validation. Here, we observed a significant inhibitory effect of the formula for strengthening the spleen and resolving blood stasis on tumor growth, and this inhibition was evaluated by observing the tumor volume. High-dose JHD showed the strongest inhibitory effect on CRC after the 5-FU-positive control when weight gain in mice was guaranteed.

Cellular senescence can be assessed by analyzing senescence markers like $p16^{Ink4a}$, lamin b1, and IL-6. The $p16^{Ink4a}$ protein plays a crucial role in governing aging, cell senescence, and the DNA damage response by restraining cell proliferation through persistent inhibition of Cdk-cyclin activity (Liu et al., 2019). Lamin b1 undergoes dynamic changes in its localization during senescence and shows decreased expression; therefore, it can be considered an indicator of senescence (Freund et al., 2012). IL-6 can also be used as a potential biomarker of cellular

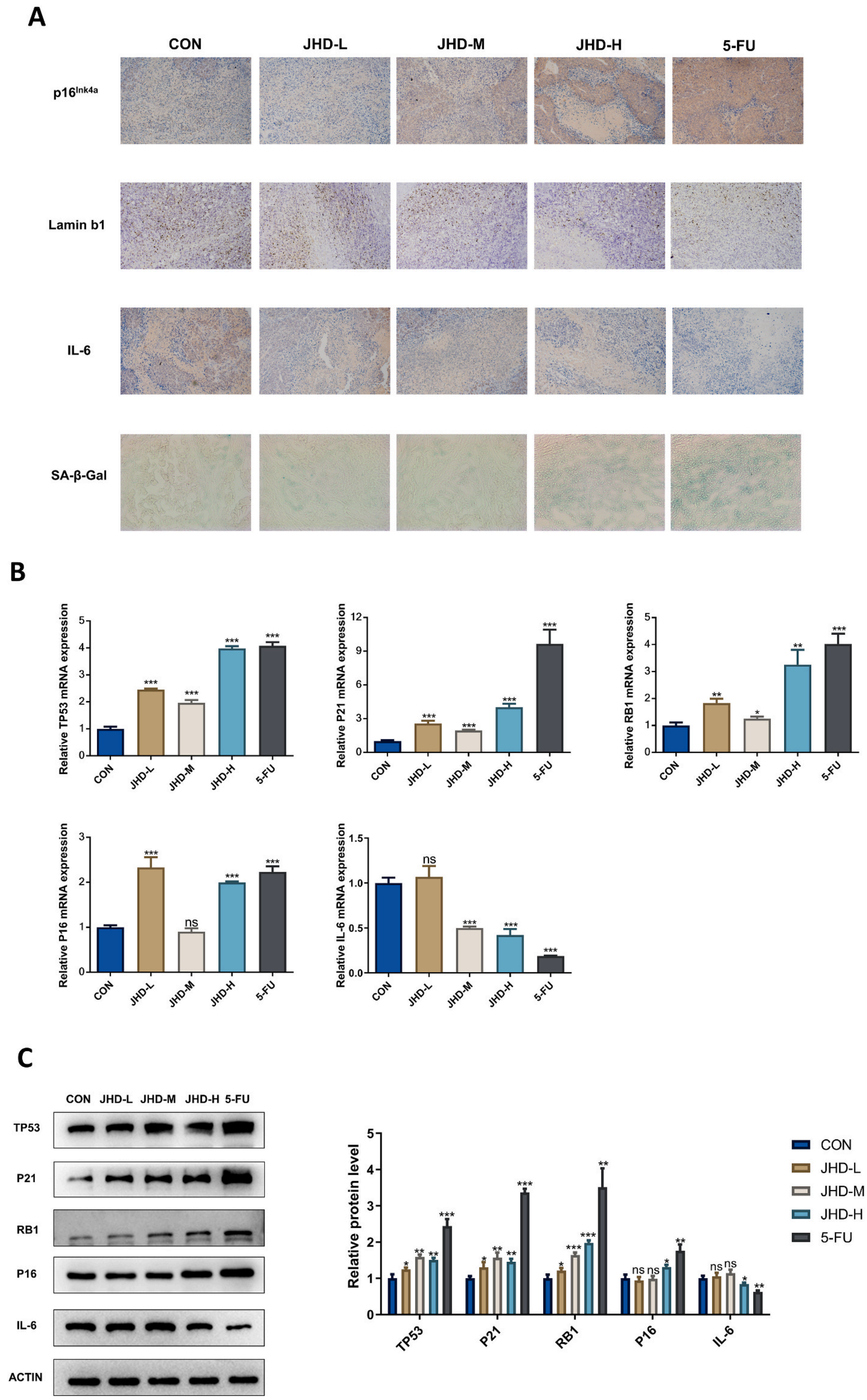

Fig. 8. JHD induced senescence and activated the p53-p21-Rb pathway in colon cancer cells in vivo. (A) Tumor sections were subjected to IHC for p16Ink4a, lamin b1 and IL-6, and SA-β-Gal staining. Original magnification was 200 × . Every single shown picture is representative of five sections from every group. Different sections represent different tumors. (B) The mRNA expression of the hub targets in rat colonic tissues, detected by qRT-PCR. (C) The protein expression of the hub targets detected by WB, and actin as internal control protein. Comparisons between means used t tests; the data are the means ± SD (*P ≤ 0.05, **P ≤ 0.001, ***P ≤ 0.0005).

senescence because of its involvement in SASPs and its association with inflammation and age-related diseases (Birch and Gil, 2020). SA-β-Gal, signifying senescence-associated β-galactosidase, stands as the most commonly employed biomarker for senescence and aging. It entails the detectable galactosidase activity in senescent cells under pH 6.0 conditions (Martinez-Zamudio et al., 2021).

In the immunohistochemistry and SA-β-Gal staining experiments, we found that JHD intervention resulted in decreased IL-6 and Lamin b1 expression, elevated p16lnk4a expression, and increased SA-β-Gal activity. The qRT-PCR findings revealed that the mRNA expression levels of TP53, P21, RB1, and P16 in the JHD groups were elevated compared to the control group. Conversely, the mRNA expression levels of IL-6 were diminished in the JHD groups in comparison to the control group. Western blot analysis indicated heightened expression of TP53, P16, P21, and RB1 proteins, along with decreased expression of IL-6 in the JHD group when contrasted with the control group. These in vivo experimental results confirm that the p53-p21-Rb signaling pathway is one of the mechanisms for inducing senescence in colon cells and that inducing cellular senescence could be regarded as one of the potential mechanisms of JHD against CRC recurrence and metastasis.

Here, we found that JHD induced cellular senescence in CRC cells, leading to tumor suppression. As anticipated through network pharmacological analysis, our attention was directed towards exploring the involvement of the p53-p21-Rb signaling pathway in the mechanistic actions of JHD against CRC (Fig. 9). Our results suggest that JHD plays a role similar to that of 5-FU in activating the p53-p21-Rb pathway and regulating IL-6 and Lamin b1 expression. Notably, the high-dose JHD group showed superior therapeutic effects to the low- and mid-dose groups in several aspects, suggesting that JHD administered at high doses may be more efficacious but this requires further validation.

This study had some limitations. Firstly, we investigated the inhibitory effect of JHD on CRC using mouse tissues. Owing to the challenges associated with acquiring human CRC cells, clinical trials have not been conducted thus far. Consequently, further efforts are necessary to bolster the clinical applicability of these findings. Secondly, there was a lack of dynamic change assessments in the p53-p21-Rb pathway in mouse CRC cells after JHD administration. Furthermore, since the tissue samples were collected after a two-week treatment duration, more in-depth investigations are warranted to unveil the extended and enduring effects of JHD treatment.

We acknowledge the potential synergies of combining JHD with conventional CRC treatments, including chemotherapy, targeted therapy, and immunotherapy. Our future research will be dedicated to developing optimal combination treatment strategies for CRC patients. These clinical trials will assess JHD's safety, efficacy, and dosage regimens, providing robust clinical evidence of its benefits. Furthermore, gaining a deeper understanding of the molecular mechanisms underlying JHD-induced cellular senescence in CRC is of paramount importance. Our ongoing research will continue to explore these mechanisms, with a specific focus on the p53-p21-Rb pathway and other associated pathways. This investigative work will establish a solid foundation for the development of targeted therapies. In addition, fostering collaboration between traditional Chinese medicine practitioners and oncologists will be vital for the successful integration of JHD into standard CRC treatment protocols. To achieve this, we are committed to actively seeking partnerships that will facilitate the seamless incorporation of JHD into clinical practice.

5. Conclusion

Here, network pharmacology, bioinformatics, and animal experiments were combined to elucidate the roles and potential targets of JHD. The intrinsic mechanism of JHD treatment in CRC is related to the induction of cellular senescence, as it interferes with the p53-p21-Rb signaling pathway and SASPs. Quercetin, isoflavones, microstegiol, and luteolin may be the key chemical compounds responsible for these effects. Overall, our findings strongly indicate that JHD holds promise as a potential candidate for CRC treatment. Moreover, these results lay the groundwork for forthcoming research endeavors aimed at delving deeper into the elucidation of JHD's mechanism of action and clinical effectiveness.

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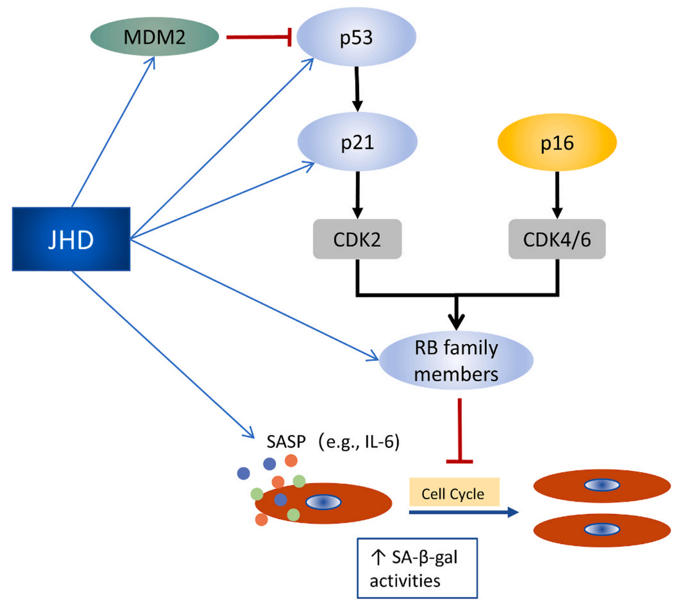

Fig. 9. Molecular mechanism of JHD acting on the p53-p21-RB signaling pathway and SASPs for the treatment of CRC.

Ethics approval of animal study

All procedures involving animals and their care in this study were undertaken in accordance with the National Institutes of Health guide for the Care and Use of Laboratory Animal, and approved by Animal Ethics Committee in Affiliated Hospital of Nanjing University of Traditional Chinese Medicine (No.2021DW-45-01).

CRediT authorship contribution statement

Shiting Wang: are the co-first author, Investigation, Data curation, Writing – original draft. **Ying Xing:** are the co-first author, Data curation. **Ruiping Wang:** Supervision, Conceptualization. **Zhichao Jin:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2023.117347.

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